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import cv2                                # Import OpenCV for image proces
import numpy as np                        # Import NumPy for matrix operat
import matplotlib.pyplot as plt          # Import Matplotlib for visualiz

# Step 1: Create a synthetic grayscale gradient image
img = np.tile(np.arange(256, dtype=np.uint8), (100, 1))

# Step 2: Simulate low quantization (4-bit = 16 gray levels)
low_quantized = (img // 16) * 16

# Step 3: Display the original and quantized images
plt.figure(figsize=(8,4))
```